

# DORAI SAI CHARAN MADISETTY

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## Education

**Amrita Vishwa Vidyapeetham, Bengaluru, India**

**2022 - Present**

*Bachelor of Technology in Computer Science with Artificial Intelligence*

*8.65 CGPA*

**Race jr college, Kurnool, India**

**2020 – 2022**

*Senior Secondary Education*

*94%*

**Montessori Indus School, Kurnool, India**

**2020**

*Secondary Education*

*90%*

## Experience

**Software Engineer Intern — New Street Tech**

**Dec 2025 – Jun 2026**

- Developed and maintained scalable RESTful APIs using Node.js for fintech applications, improving backend service efficiency and reliability.
- Contributed to a microservices-based architecture, developing and integrating distributed backend services for enhanced system scalability.
- Implemented backend features following software engineering best practices, including code quality, modular design.
- Designed, optimized, and executed complex PostgreSQL queries to ensure efficient data retrieval and DB performance.

## Projects

**Multimodal AI Research Assistant**

**GITHUB**

- **Tech:** Python, FastAPI, Next.js, TypeScript, ChromaDB, LangChain, EasyOCR, Sentence-Transformers, PyTorch
- Built a Multimodal RAG system indexing research PDFs by content type into ChromaDB using embeddings, with 9 task-specific pipelines.
- Implemented a ReAct agentic reasoning framework for multi-hop reasoning across papers, and a dual-LLM architecture for generation and Gemini 2.0 Vision for figure description and LaTeX equation extraction
- Developed an AI detection and humanization engine using RoBERTa blended with heuristic scoring. FastAPI backend with 14 REST endpoints.

**AI-Driven Load Balancer for Cloud Computing Environments**

- Developed a reinforcement learning-based load balancing system using **Deep Deterministic Policy Gradient (DDPG)** to optimize CPU utilization and minimize task latency across a 5-node AWS EC2 cluster
- Reduced memory usage by 14% and energy consumption by 15% compared to traditional heuristic-based load balancing approaches
- Applied concepts of distributed systems, cloud computing, and performance optimization in a real-world environment
- Published research at the **2025 International Conference on ICT for Sustainable Development(ICT4SD)**, in ICT Analysis and Applications (Lecture Notes in Networks and Systems)

**FINARC – Financial Intelligence & Transaction Management System**

**GITHUB**

- **Tech:** Python, FastAPI, Next.js, TypeScript, PostgreSQL, SQLAlchemy, JWT
- Developed scalable **RESTful APIs** using FastAPI with 30+ endpoints for transaction processing, account management, and financial analytics
- Designed and optimized **PostgreSQL database schema using SQLAlchemy ORM**, improving query performance and enabling efficient handling of multi-account ledgers and recurring transactions

## Skills

**Programming Languages:** C++, Python, SQL, JAVA

**Backend Development:** FastAPI, REST APIs, JWT Authentication, PostgreSQL, Redis

**Tools & Technologies:** Git, GitHub, Linux

**Machine Learning & Gen AI:** Transformers, RAG, LangChain, Prompt Engineering, Numpy, Pandas, Scikit-learn, TensorFlow, PyTorch, Matplotlib

## Research Publications

**Solar Panel Fault Detection: Comparative Analysis of DL models**

**IEEE Paper**

- Published at 2025 IEEE 2nd International Conference on Information Technology, Electronics and Intelligent Communication Systems (ICITEICS)

## Certifications

- AWS Academy Graduate – Introduction to Cloud Computing [Source](#)
- Image Processing Onramp - MATLAB [Source](#)
- Computer Vision Onramp -MATLAB [Source](#)